

TRIP-Sense: A Dynamic Travel Planner Using NLP and Hybrid Machine Learning Models

Abstract

Modern travel planning often relies on static recommendations that fail to adapt to user behaviour or dynamic real-world contexts. TRIP-Sense - short for Travel Recommendation and Intelligent Planning - addresses this gap by introducing an AI-driven travel planner that learns, senses, and personalizes each journey. The system integrates Natural Language Processing (NLP) and Hybrid Machine Learning models to generate itineraries that evolve with changing environmental factors. The NLP module interprets user intent from natural language queries and performs sentiment analysis on reviews to capture nuanced traveller interests. A hybrid recommendation engine combining Collaborative Filtering, Content-Based Filtering, and SVD - KNN stacking generates personalized destination and activity suggestions. Real-time data from APIs such as Google Maps, OpenWeather, and Eventbrite enable the system to adapt intelligently - for instance, recommending indoor alternatives during unfavourable weather. Graph-based route optimization (Dijkstra/A*) ensures time- and cost-efficient planning. An interactive Streamlit dashboard visualizes destinations, routes, and expenses dynamically. By merging contextual awareness, adaptive intelligence, and seamless visualization, TRIP-Sense transforms itinerary generation into a living, learning travel ecosystem, making every trip uniquely personal and context-aware.

Keywords

NLP, Hybrid Machine Learning, Travel Recommendation, Context Awareness, Sentiment Analysis, Itinerary Optimization, Adaptive System.

Team:

Manaswi Priya Maddu

Praneetha Gorli

Pavan Kumar Bora

Hari Krishna Neerasa

Guide:

Dr. K. Chaitanya